

Motion-Based Salient Moment Extraction from 3D VR Training Records for Reflective Learning

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ABSTRACT

Virtual Reality (VR) training sessions can be recorded and replayed, enabling immersive after-action review (AAR). However, long-duration VR recordings are costly to review due to combined temporal and spatial exploration overhead. We propose a motion-based salient moment extraction method that operates directly on 3D VR recordings without predefined event logs. The method analyzes object-level transform changes across the scene and extracts representative moments via curve simplification and density-based temporal aggregation. Comparisons with 2D first-person-view video summarization show that the proposed 3D method achieves higher recall and robustness, demonstrating its suitability for efficient VR-based AAR.

Index Terms: Virtual Reality, After-Action Review, Salient Moment Extraction, 3D Scene Analysis, Motion Analysis

1 INTRODUCTION

Unlike real-world experiences, Virtual Reality (VR) experiences can be recorded and replayed, allowing users to relive past interactions when appropriate logging or recording is available. This makes VR particularly well suited for education and training, as learners can revisit their actions and decisions after completing a task via after-action review (AAR). However, long-duration VR recordings introduce significant challenges; reviewers must identify when and where meaningful events occur within the 3D scene. Consequently, there is a growing need for automated techniques that can identify salient moments and summarize lengthy recordings to improve feedback efficiency [1].

To address this, recent studies have proposed methods for automatically labeling or summarizing user sessions, such as generating Large Language Model (LLM)-assisted analytic insights from recorded interaction data [2] or supporting collaborative annotation and query-based retrieval in replayed social VR sessions [3]. However, these approaches typically rely on rule-based event logs, requiring developers to predefine key events and manually instrument scenarios for each piece of content. Such manual specification imposes substantial development overhead, limits scalability, and fails to capture undefined or emergent behaviors.

Alternatively, prior research has explored identifying representative keyframes from user tracking data without predefined event definitions. For example, motion keyframes can be obtained by clustering head and hand movements and taking each cluster's median timestamp [4]. However, these approaches primarily reflect the user's motion and often fail to capture meaningful changes in the broader scene context. If environmental dynamics change (e.g., a door opening or a non-player character (NPC) appearing) while the user remains relatively static, these methods may incorrectly treat the event as insignificant.

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To overcome these limitations, we propose a scene-aware framework that captures salient moments based on state changes across the entire virtual environment. By leveraging 3D scene data, our approach identifies salient moments without predefined event definitions or user-centric assumptions. We validate our method by comparing salient moments extracted from 3D scene data with those obtained from 2D first-person-view (FPV) recordings, demonstrating its effectiveness in capturing environmentally meaningful events. Our method is better suited for VR-based AAR, as it preserves scene structure and supports in-context interaction.

2 PROPOSED METHOD

Below, we describe our method for extracting salient moments from VR recordings. The input comprises time-series position (P_t) and rotation (Q_t) data for all moving objects in the scene hierarchy.

$$C_{trans}(t) = \|P_t - P_{t-1}\|^2, \quad C_{rot}(t) = 2 \arccos(|Q_t \cdot Q_{t-1}|)^2 \quad (1)$$

To quantify changes, we compute the squared L_2 norm of the inter-frame translation and the squared angular difference between quaternions, as defined in Equation 1. For each object, we apply the Discrete Curve Evolution (DCE) algorithm to simplify its motion curve in the (t, y) domain [5].

$$S(u, v, w) = \frac{1}{2} |u_t(v_y - w_y) + v_t(w_y - u_y) + w_t(u_y - v_y)| \quad (2)$$

In Equation 2, the vertices u, v, w represent points in the (t, y) domain, where t corresponds to the frame index and y to the motion magnitude. DCE assigns a relevance cost to each interior vertex v using the triangle area with its neighbors u and w . Vertices with the smallest cost are iteratively removed, which suppresses high-frequency tracking jitter while retaining shape-defining transitions. We stop when the next removal cost exceeds an adaptive cutoff $S_{opt} = \mu_S - \log(\gamma)\sigma_S$ (with $\gamma = 0.8$), where μ_S and σ_S are the mean and standard deviation computed from the recorded removal costs. Finally, we select the local minima of the simplified curve as local candidates to capture motion boundaries, indicating transitions between inactive and active motion segments. An avatar, a building, or an object typically consists of multiple descendant objects. Aggregating local candidates from all objects can therefore yield an excessive number of candidates.

$$D(t) = \sum_{i \in \text{Group}} 1(t \in S_{local}^{(i)}) \quad (3)$$

To address this, we project all local candidates onto a single timeline and construct the timeline density function $D(t)$, as defined in Equation 3. Here, $S_{(i)local}$ denotes the set of local candidate timestamps identified for the i -th object. $D(t)$ counts the number of objects that have a local candidate at time t . We apply DCE to $D(t)$ and extract its local maxima as global candidate event timestamps.

Density-based candidate extraction may detect a single event across multiple temporally adjacent frames. To mitigate this, the object activity pattern at each candidate time t is represented as a binary motion status vector V_t . Cosine similarity $Sim(t_a, t_b)$ is computed for candidate pairs within a 1-second temporal. Pairs with $Sim(t_a, t_b) \geq 0.7$ are merged into a single event, ensuring that temporally close and kinematically similar candidates are merged.

3 PRELIMINARY STUDY

3.1 Experimental Setup

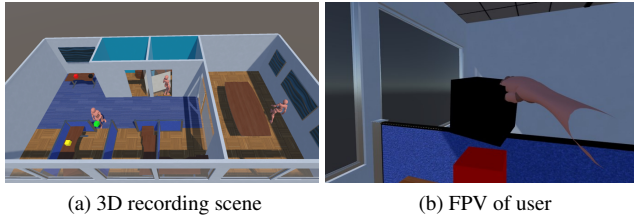


Figure 1: Experimental setup and observation views of the same event: (a) 3D recording scene, (b) first-person view of the user.

For the experiment, a virtual environment was constructed and recorded using a 3D recording system [3], as illustrated in Figure 1. The environment included a user with full-body tracking, two NPCs performing state-based animations, a dynamic door, and four interactive cubes. To evaluate detection accuracy, ground truth was defined based on logged timestamps of actual state changes. Specifically, ground-truth events were categorized as follows: NPCs (start and end points of transitions between *Idle*, *Walk*, and *Sit* animations), the door (four stages: *Opening Start*, *Fully Open*, *Closing Start*, and *Fully Closed*), and cubes (*Grab* and *Release* interaction moments). The recording duration was 83 s, yielding 42 ground-truth event timestamps. Salient moments extracted within ± 0.5 s of a ground-truth timestamp were considered correct.

3.2 Comparative Methods and Post-processing

We validated the proposed 3D salient moment extraction method by comparing it with a 2D video-based baseline. The 2D method employed Video SUMMARization (VSUMM) [6], which clusters frames based on visual features such as color histograms. VSUMM was applied to FPV videos rendered from the 3D recordings to extract salient moments reflecting the user’s visual experience. To mitigate the limitation of a single viewpoint, additional FPV videos from NPCs were also included.

In contrast, the proposed 3D method extracted salient moments by analyzing object-level state changes within the scene. Since both methods produced salient moment sets from different sources, we merged the results within each method and grouped detections within a 0.25 s window to avoid temporal redundancy. As both approaches rely on frame-to-frame changes, we evaluated performance at 30 and 60 fps to examine robustness across sampling rates.

3.3 Result and Discussion

Table 1: Comparison between 3D and 2D Methods

FPS	Method	Accuracy	Precision
60fps	3D method	97.6% (41/42)	55.4% (41/74)
	2D method	85.7% (36/42)	39.1% (36/92)
30fps	3D method	85.7% (36/42)	54.5% (36/66)
	2D method	52.3% (22/42)	48.9% (22/45)

Table 1 presents a performance comparison between the proposed 3D method and the 2D method. Our 3D method consistently outperformed the 2D method across all metrics. At 60fps, the 3D method achieved a 97.6% recall, detecting all but one event, whereas the 2D method achieved only 85.7%. The 3D method showed superior robustness. As shown at 30fps, while the 2D method’s recall dropped to 52.3%, the 3D method maintained 85.7%. Furthermore, high-precision results indicated that the 3D method provides a more concise summary of salient moments.

A notable limitation of the 2D method was its tendency to detect salient moment corresponding to objects outside the camera’s field of view, caused by coincidental visual similarities during multi-view data integration. Filtering out these reduces accuracy to 45.2% (60fps) and 33.3% (30fps). However, the 3D method remained unaffected by this issue as it relies directly on the object’s transform. This suggests that the 3D method may exhibit higher practical performance than indicated in Table 1.

An analysis of error cases observed in the 3D method under both 60 fps and 30 fps conditions revealed a discrepancy in character movement. While the system log indicated a transition from ‘Walk’ to ‘Idle’, the animation visually ceased before the corresponding log timestamp. During cube interactions, the 30 fps configuration prioritized the ‘Moment of complete stop’ over the ‘Release’ event. In contrast, the 60 fps configuration leveraged its higher temporal resolution to successfully capture both instances. These findings highlight the distinctive characteristics of the proposed method relative to the limitations of pre-defined logging systems. Such systems often fail to record undefined physical interactions (e.g., falling objects) or subtle visual discrepancies. Conversely, the experimental results demonstrate that the proposed method effectively captures actual movements and sudden state changes within the content without relying on pre-defined scenarios.

4 CONCLUSION AND FUTURE WORK

In this paper, we proposed a salient moment extraction method utilizing 3D object transform data and validated it by comparing it with VSUMM, a 2D video summarization algorithm. The experimental results indicate that the proposed method achieved higher Recall and Precision than the 2D baseline. By directly leveraging transform information, our method mitigated field-of-view constraints. Building on these results, we will conduct additional tests on more diverse datasets and scenarios to further evaluate robustness and generalizability. We also plan to develop an intent-aware filtering module that analyzes user queries over the extracted object-specific salient moment, providing only contextually relevant information to improve analysis efficiency.

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